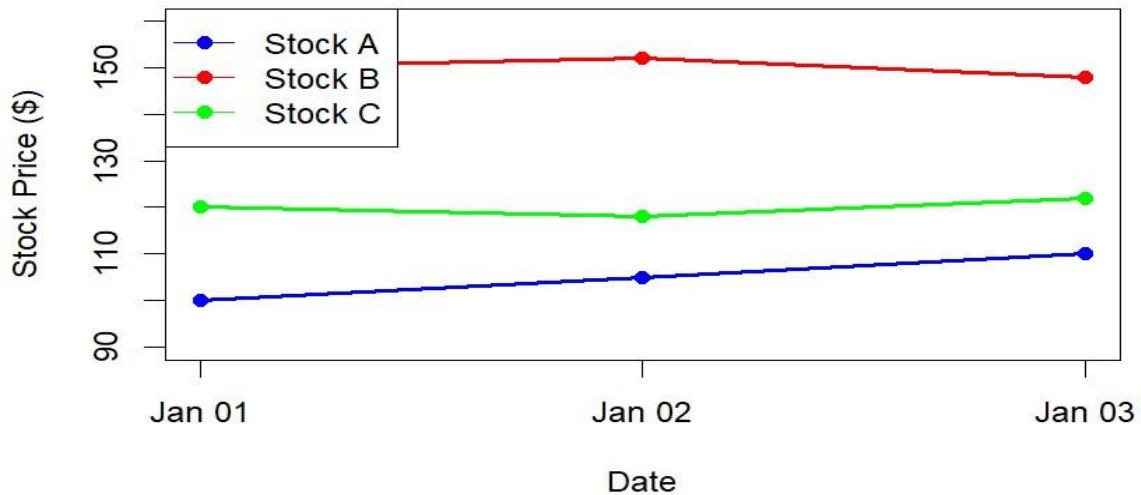


SET 16

QUESTION-1

Stock Price Analysis



CODE:

```
# Stock Price Data
date <- as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03"))
stock_A <- c(100, 105, 110)
stock_B <- c(150, 152, 148)
stock_C <- c(120, 118, 122)

# Line Chart
plot(date, stock_A, stock_B, stock_C,
      type = "o",
      col = "blue",
```

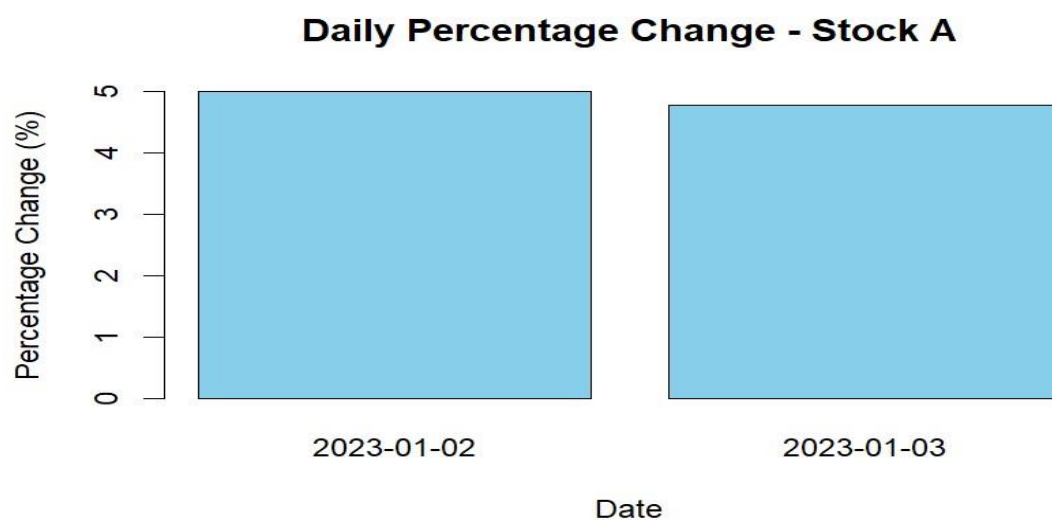
```

pch = 16,
lwd = 2,

    xlab = "Date",    ylab = "Stock
Price ($)",    main = "Stock Price
Analysis",    ylim = c(90, 160))
lines(date, stock_B,    type = "o",
col = "red",    pch = 16,    lwd =
2) lines(date, stock_C,    type =
"o",    col = "green",    pch = 16,
lwd = 2) legend("topleft",    legend
= c("Stock A", "Stock B", "Stock C"),
col = c("blue", "red", "green"),    lty
= 1,    pch = 16)

```

QUESTION-2



CODE:

```

# Stock A Prices stock_A
<- c(100, 105, 110) #

```

Calculate Daily

Percentage Change

```
percentage_change <- c(  
  NA,  
  ((stock_A[2] - stock_A[1]) /  
stock_A[1]) * 100,  
  ((stock_A[3] - stock_A[2]) /  
stock_A[2]) * 100  
)
```

```
print(percentage_change) #
```

Remove first NA value

```
change <-
```

```
percentage_change[2:3]
```

```
days <- c("2023-01-02",
```

```
"2023-01-03")
```

```
# Bar Chart barplot(change,
```

```
names.arg = days,        col =
```

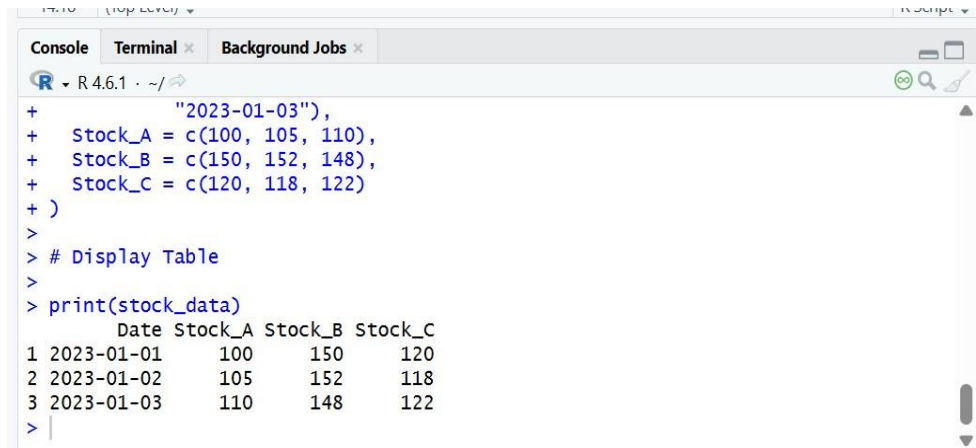
```
"skyblue",        xlab = "Date",
```

```
ylab = "Percentage Change (%)",
```

```
main = "Daily Percentage Change -
```

```
Stock A")
```

QUESTION-3



```
R 4.6.1 · ~/
+       "2023-01-03"),
+   Stock_A = c(100, 105, 110),
+   Stock_B = c(150, 152, 148),
+   Stock_C = c(120, 118, 122)
+ )
>
> # Display Table
>
> print(stock_data)
      Date Stock_A Stock_B Stock_C
1 2023-01-01     100     150     120
2 2023-01-02     105     152     118
3 2023-01-03     110     148     122
>
```

CODE:

```
# Stock Price
```

```
Dataset
```

```
stock_data <-
```

```
data.frame(
```

```
  Date = c("2023-
```

```
01-01",
```

```
          "2023-01-02",
```

```
          "2023-01-03"),
```

```
  Stock_A = c(100, 105,
```

```
110), Stock_B = c(150,
```

```
152, 148),
```

```
  Stock_C = c(120, 118, 122)
```

```
)
```

```
# Display Table
```

```
print(stock_data)
```